1 Exactly how those rewards are doled out, and to whom, is not the part they have in common.

2 This view is also espoused, though not explored in philosophical detail, in the classic early-2000s TV series *Battlestar Galactica.*

3 apocalypsi?

4 We do this by bouncing it. Really. First we cool the neutrons to almost absolute zero, then we slow them to jogging speed, then we bounce them up and down like a Ping-Pong ball on a paddle. And this also tells us something about dark energy, the mysterious something that makes our whole universe expand faster. Physics is wild.

5 String theorists produce a lot of these theories. (String theory is a blanket term for theories that try to bring together gravity and particle physics in new ways, but most of the work done to develop it now relies on mathematical analogs rather than anything pertaining to the "real" world.) Sometimes when I'm in string theory talks, I have to resist the urge to raise my hand and clarify that none of these calculations pertain to *our* universe, just in case anyone in the room is as confused as I first was when I started attending string theory talks.

6 This is, of course, one of the most fun things I've ever worked on, hence this book. I'm not sure why I like it so much. It may be a bad sign.

7 And we're talking about the universe here, so I really do mean EVERYTHING.

8 Please stick around until Chapter 4, when the Boltzmann Brain community will get their proper due.

9 Technically it is called a "bubble of true vacuum," which, to be fair, also sounds pretty darn ominous.

10 Another spoiler: it's not looking great.